

Performance Metrics Classification Workshop

Contributors:

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Project Repository:

<https://github.com/alicih4n/PerformanceMetricsClassification.git>

Project Summary:

In this workshop, we successfully implemented a binary classification system for the MNIST dataset. We trained a Stochastic Gradient Descent (SGD) classifier to distinguish the digit '5' from all other digits. We explored various performance metrics, demonstrating that Accuracy is misleading for imbalanced datasets (where '5's are rare). We implemented and analyzed Confusion Matrices, Precision, Recall, and F1 Scores to provide a more robust evaluation. Finally, we compared our model against a Random Forest classifier using ROC Curves and AUC scores, visualizing the trade-offs between sensitivity and specificity.